

1. MACHINE IDENTIFICATION

The Press-2001 is a 1500-ton mechanical (flywheel) press with serial prefix MP-. Unlike the Press-2000, this machine has NO hydraulic system. Clamping force is generated by a motor-driven flywheel and eccentric gear mechanism. Do NOT apply Press-2000 hydraulic error codes or corrective actions to this machine. The Press-2001 uses a pneumatically actuated clutch and brake unit with a spring-applied, air-released brake design.

1.1 LUBRICATION SYSTEM

The main gearbox and clutch assembly are splash-lubricated with ISO-VG220 gear oil. The reservoir holds 6 L. Change the oil every 3000 operating hours or 12 months, whichever comes first. The slide guides are grease-lubricated via grease fittings on the front and rear of each guide rail. Apply lithium-based NLGI 2 grease weekly.

1.2 CLUTCH/BRAKE UNIT

The press uses a pneumatically actuated clutch and brake unit. Air supply must be 6-8 bar, dry, and filtered. The clutch engages when the solenoid valve opens, releasing the spring-applied brake. Brake wear is indicated by increased stopping time. Measure the stopping angle annually ? the crankshaft should stop within 30° of rotation from the clutch release point.

1.3 FLYWHEEL SPECIFICATIONS

Flywheel diameter: 1.8 m

Flywheel speed: 180 RPM at 60 Hz

Main motor: 55 kW, 480V 3-phase, 1200 RPM base speed

VFD: Allen-Bradley PowerFlex 525, 75 kW

Belt drive: 8V-2650, 6 belts

7. TROUBLESHOOTING

7.1 E101 ? Drive Chain Binding

CRITICAL: E101 on the Press-2001 means DRIVE CHAIN BINDING. It does NOT mean winding overtemperature (RoboInject-300) or low hydraulic pressure (Press-2000). Always verify the machine model before diagnosing.

Meaning: The main drive chain tension is excessive, or the chain is catching on the chain guard. The controller detects increased torque demand on the flywheel motor via the VFD current monitoring circuit.

Probable causes:

1. Drive chain tension too high (overtightened) ? the chain should have 15-20 mm of vertical deflection at the center of the longest span when pressed with 5 kg force.
2. Chain sprocket misalignment ? the motor sprocket and flywheel sprocket must be aligned within ± 1 mm.
3. Worn chain links causing the chain to skip under load ? inspect for stiff links, stretched pins, or uneven wear patterns.
4. Foreign object caught in the chain guard ? check for tools, debris, or rags that may have fallen into the chain enclosure.
5. Flywheel bearing seizure ? indicated by excessive heat or noise near the flywheel bearing housings.

Corrective action:

Step 1. Remove the chain guard (12 $\times$ M10 bolts). Inspect the chain visually for damage, stiff links, or foreign objects.

Step 2. Measure chain tension at the center of the longest span. Using a spring scale, apply 5 kg of force to the chain. The deflection should be 15-20 mm. Adjust by loosening the motor mounting plate bolts and moving the plate on the slide rails.

Step 3. Check sprocket alignment using a straightedge placed across both sprockets. Misalignment greater than 2 mm causes accelerated chain and sprocket wear. Shim the motor mounting plate as needed.

Step 4. Lubricate the chain with ISO-VG320 gear oil applied evenly across the full chain length. Run the motor without load for 5 minutes.

Step 5. If binding persists, measure chain pitch. If the pitch has increased by more than 2% from the original specification, replace the chain set (P/N CH-2001-KIT includes chain and both sprockets).

Source: Section 7, page 311.

7.2 E204 ? Flywheel Overspeed

Meaning: The flywheel speed has exceeded 110% of rated maximum (198 RPM). The VFD has applied the dynamic brake resistor to slow the flywheel.

Probable cause: VFD parameters misconfigured or feedback encoder fault. Check VFD parameter P-04 (Maximum Frequency) ? should be 60 Hz for 180 RPM.

7.3 E312 ? Clutch Air Pressure Low

Meaning: The air supply to the clutch solenoid valve has dropped below 5 bar. The press controller will not allow a cycle to start while the pressure is low.

Corrective action:

Step 1. Check the main air supply pressure at the regulator adjacent to the press. The regulator should be set to 6-8 bar.

Step 2. Inspect the air line filter/water separator. Drain any accumulated water by opening the petcock at the bottom of the bowl.

Step 3. Test the solenoid valve by applying 24V DC directly to the valve coil terminals. If the valve clicks but no air flows, the valve spool may be stuck. Remove and clean the spool. If the valve does not click, replace the coil (P/N V-SOL-2001).

8. PERIODIC MAINTENANCE

Daily: visually check chain tension, listen for abnormal gear noise, verify air pressure on the regulator gauge, check that the flywheel reaches full speed within 30 seconds of start.

Weekly: lubricate the slide guide rails via grease fittings, drain the air line filter, visually inspect the clutch brake pads for wear.

Monthly: check gearbox oil level, inspect flywheel bearing temperature by touch (should be cool to the touch after 30 minutes of operation), check belt tension and alignment.

Annually: replace gearbox oil (ISO-VG220), replace clutch brake pads, perform a full electrical inspection of the VFD and motor windings, measure and record chain pitch for wear tracking.